```
class CPUNonBoundedSpec ... {
  ...

  "calling to a real service" should {
    "be handle with throttle and mapAsync" in {

      implicit val validatedFormat
          : RootJsonFormat[HttpServer.Validated] =
        jsonFormat1(HttpServer.Validated)                ❶

      val result: Future[Done] =
        Source(1 to 100)
          .throttle(32, 1.second)                        ❷
          .mapAsync(100)(each =>                         ❸
            Http().singleRequest(
              Get("http://localhost:8080/validate?quantity=" + each)))
          .mapAsync(100)(each =>
            Unmarshal(each).to[HttpServer.Validated])    ❹
          .run()

      Await.result(result, 5.seconds)
    }
  }
  ...
}
```

❶ RootJsonFormat used later to unmarshal

❷ Throttles to the default max-open-requests in Akka HTTP (that is, 32)

❸ Asynchronously calls the HTTP service

❹ Asynchronously transforms each response into Validated

To run this test, enter the following:

```
sbt "chapter16/testOnly *SamplesSpec -- -z \"calling to a real service\""
```

Connecting to an external database

Normally you need to enrich the data you are processing. However, calling a service that, in turn, calls a database may cause too much latency. Let’s see how to call the database directly.